MUHAMMAD AZIQ RAUF

AI / ML Engineer · LLM Systems, RAG & Speech AI
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SUMMARY

AI engineer who has built LLM and speech systems for hospitals in Bahrain (AI scribe, PHI masking, bilingual voice agent). Builds RAG pipelines, fine-tunes LLMs with LoRA, and ships models with MLOps and evals. Based in Bahrain, available immediately.

EXPERIENCE

MedPro **Sep 2026 – Nov 2026**
AI Engineer Intern — Hospital Management System *Manama, Bahrain*

- Built an AI scribe for a hospital management system used by hospitals in Bahrain: Whisper transcribes doctor–patient consultations and the Claude API auto-fills structured clinical forms from the transcript, reducing manual charting for doctors.
- Designed a PHI/PII masking layer that strips patient identifiers before any LLM call, so no identifiable patient data reaches third-party APIs.
- Benchmarked self-hosted speech-to-text against API models on word error rate across Arabic, South Asian and native English accents, and on cost, to guide the self-hosted vs API decision for production.
- Built an inbound voice AI agent (n8n + ElevenLabs) on a Zain Bahrain line that serves callers in English and Arabic, integrates with the HMS API, database and calendar, and hands off to a human agent when needed.

Ben Da'ood Studios **Jun 2026 – Aug 2026**
AI Automation Engineer *Islamabad, PK*

- Built end-to-end n8n workflows automating website content publishing and internal operations for a creative agency.
- Replaced repetitive manual processes with automated pipelines the team runs daily, freeing time for client work.

Cloudtek **Jun 2025 – Aug 2025**
AI Developer Intern *Islamabad, PK*

- Developed TensorFlow models and integrated them into production, user-facing web applications.
- Cleaned and preprocessed complex datasets, speeding up the model training pipeline by 30%.

PROJECTS

LLM-Orchestrator-RAG-Server | *Django, PostgreSQL/pgvector, Celery/Redis, Langfuse*

- Architected a multi-user RAG server (per-user data scoping) with hybrid retrieval (pgvector dense search + Postgres full-text via reciprocal rank fusion) and cross-encoder reranking, lifting MRR from 0.80 to 0.92 on a held-out eval set.
- Built an async ingestion pipeline (Celery, Redis) with Langfuse tracing for cost and latency, backed by 56 automated tests.

Q&Ace: Multimodal AI Interview Coach | *Python, Llama 3.1/3.3, LoRA (MLX), Whisper, React*

- Engineered an interview coach combining facial emotion recognition, Whisper speech-to-text and LLM feedback: Groq-hosted Llama 3.3 70B for low-latency inference plus two LoRA adapters fine-tuned locally on Llama 3.1 for coach and evaluator roles.
- Reduced response latency with asynchronous processing, running 3 models concurrently without UI lag.

AegisVision: Real-Time Traffic Monitoring | *YOLOv11n, ByteTrack, ONNX, CoreML, Docker*

- Built real-time vehicle detection and bidirectional tracking (YOLOv11n + ByteTrack) with JSONL telemetry per class.
- Reached 47 FPS at 640×640 on Apple M1 Max; exported to ONNX for headless edge deployment via Docker.

PhoenixML: Automated MLOps Fraud Detection | *Python, MLflow, FastAPI, Evidently, GitHub Actions*

- Automated retraining with Evidently drift detection: models register to MLflow Staging, with human-gated promotion.
- Served the fraud model (PR-AUC 0.84 on highly imbalanced data) via Dockerized FastAPI on GHCR; CI runs 36 pytest tests.

SKILLS

Programming: Python, SQL, JavaScript, Java, C++
AI/ML: LLM agents & tool calling, RAG, LLM fine-tuning (LoRA), speech-to-text, computer vision, MLOps, evals & tracing
Frameworks & Tools: PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face, Claude & OpenAI APIs, n8n, ElevenLabs
Infrastructure: Docker, FastAPI, Django, PostgreSQL/pgvector, Celery/Redis, MLflow, Langfuse, Git, GitHub Actions, AWS
Spoken Languages: English, Urdu

EDUCATION

FAST-NUCES (National University of Computer and Emerging Sciences) **2026**
Bachelor of Science in Software Engineering *Islamabad, PK*

- **Top 5% nationally** (of 50,000+ final-year students) in the HEC National Skill Competency Test.
- **Certifications:** IBM AI Engineering Professional Certificate; IBM LLM Development Certificate.

LEADERSHIP & PUBLICATIONS

- Led a 4-person team at the NUST Hackathon (Feb 2026); co-authored a SciNex Journal of Advanced Sciences paper (2025).